

1. Executive Summary & Architecture

StockFlow is an enterprise inventory management and shift auditing system engineered for high-accuracy medical and commercial environments. It automates inventory tracking, shift handovers, financial valuation, and security auditing.

SYSTEM ARCHITECTURE & TECH STACK

FRONTEND ARCHITECTURE

Built with Vanilla HTML5, ES6+ JavaScript, and CSS3. Operates dependency-free for maximum browser speed, responsive Tailwind styling, and live client data filters.

BACKEND REST API

Powered by Node.js & Express.js v5. Implements stateless JWT authentication, password hashing with bcryptjs, parameterized SQL protection, and audit logging.

CORE COMPONENT SCHEMA

MODULE	TECHNOLOGY	KEY RESPONSIBILITY	SECURITY CONTROL
Auth Engine	JWT + Bcrypt	User login, token issuance, password validation	RBAC Admin Guard
Inventory Module	Express + SQL	Stock entries, opening/closing calculations	Ownership Check
Financial Engine	PostgreSQL CTE	Stock valuation (In, Out, Damaged, Current)	Admin Only Rights
PDF Renderer	Offscreen Iframe	Compiled client-side PDF document generation	Isolated Context

SECURITY & ROLE-BASED ACCESS CONTROL (RBAC)

Standard Staff User

- Record daily morning and night shift stock counts.
- View active product catalogue & self-created entries.
- Restricted from editing entries created by others.

System Administrator

- Full CRUD permissions across Products, Users, & Stock.
- Access to Audit Logs, Financial Valuations & Loss Reports.
- Database auto-recovery & system diagnostics rights.

2. Shift Operations & User Guide

StockFlow streamlines daily inventory workflows by automatically bridging morning closing stock with night shift opening stock, preventing unrecorded inventory leakage.

MANDATORY SHIFT HANDOVER LOGIC

Expected Stock = Opening + Received - Stock Out - Damaged

Variance = Physical Closing Count - Expected Stock

```
// Automated Shift Handover Algorithm (inventoryController.js)
if (shift === "night") {
  const morning = await getMorningClosing(product_id, date);
  if (morning) openingStock = morning.closing; // Auto-link Opening
  variance = physicalClosing - (openingStock + received - damaged - disbursed);
}
```

DAILY OPERATIONAL STEPS

- 1. Authentication & Shift Selection**

Staff log in using assigned credentials. Session shift (Morning or Night) and active working date are established.
- 2. Recording Product Stock**

Staff input physical counts for Received, Stock Out, Damaged, and Closing Count. System alerts on variance.
- 3. Catalogue Management (Admin)**

Admins manage inventory items. Product names are normalized (lower(name) deduplication) to prevent duplicate listings.

INTERFACE DESIGN & RESPONSIVENESS

- Desktop View: High-density data grid with sticky header rows for fast scrolling during bulk shift audits.
- Mobile View: Touch-optimized card list with bounded containers (max-h-[60vh] overflow-y-auto).

3. Financial Reporting & Maintenance

StockFlow provides real-time financial reporting for management alongside an append-only audit trail and automated database disaster recovery options.

FINANCIAL STOCK VALUE SUMMARY METRICS

TOTAL RECEIVED Received Value RWF	TOTAL EXITED Stock Out Value RWF	DAMAGED LOSS Financial Loss RWF	VALUATION Current Value RWF
--------------------------------------	-------------------------------------	------------------------------------	--------------------------------

ISOLATED PDF GENERATION ENGINE

- PDF reports use an isolated offscreen rendering engine (exportHTMLToPDF) to guarantee zero canvas clipping or blank pages:
- Offscreen Iframe Isolation: Mounts content in a hidden iframe (1040px landscape / 740px portrait) to bypass page CSS.
 - Base64 Asset Embedding: Injects high-resolution DMC Hospital Logo directly via inline Base64 data strings.
 - Instant Preview & Download: Generates a downloadable .pdf file and opens a viewable browser blob tab.

AUDIT TRAIL & DATABASE AUTO-RECOVERY

All system actions (CREATE, UPDATE, DELETE) write to an immutable audit_logs table containing previous and updated JSON snapshots. Setting AUTO_RESTORE=true enables automated disaster recovery from audit logs.

ENVIRONMENT SETUP GUIDE

```
# backend/.env Configuration
PORT=5000
NODE_ENV=production
DATABASE_URL=postgresql://user:password@host:5432/stockflow
JWT_SECRET=your_super_secret_jwt_key
JWT_EXPIRES_IN=8h
AUTO_RESTORE=false
```


